

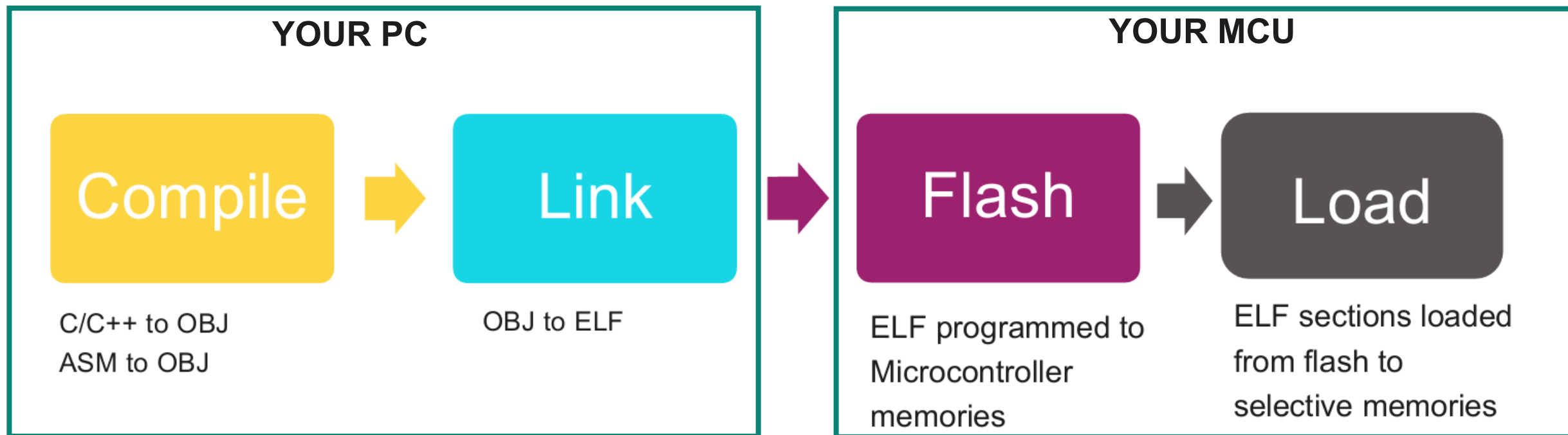


Embedded System Course – OpenOCD and GDB

Vinay Kumar Boya
18/Oct/2025

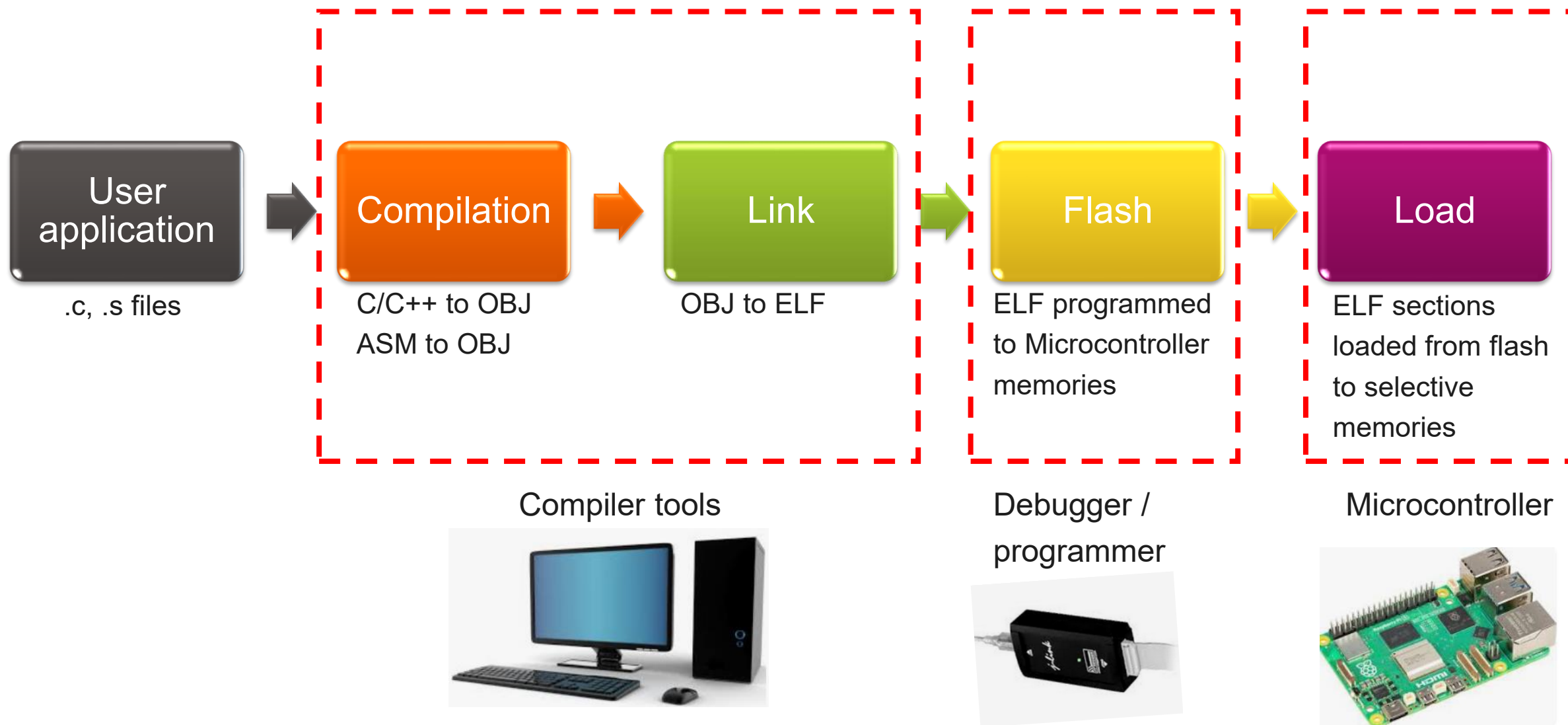


Source to binary to Flash – Flow (RECAP)

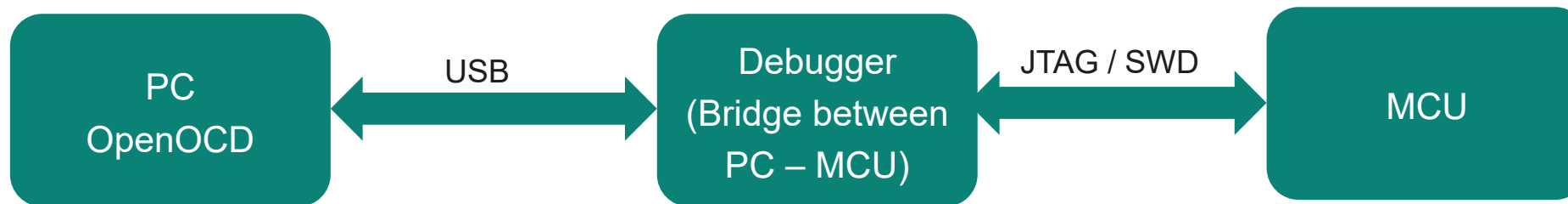


Who is Bridge?

Source to binary - Flow



Program and Debug



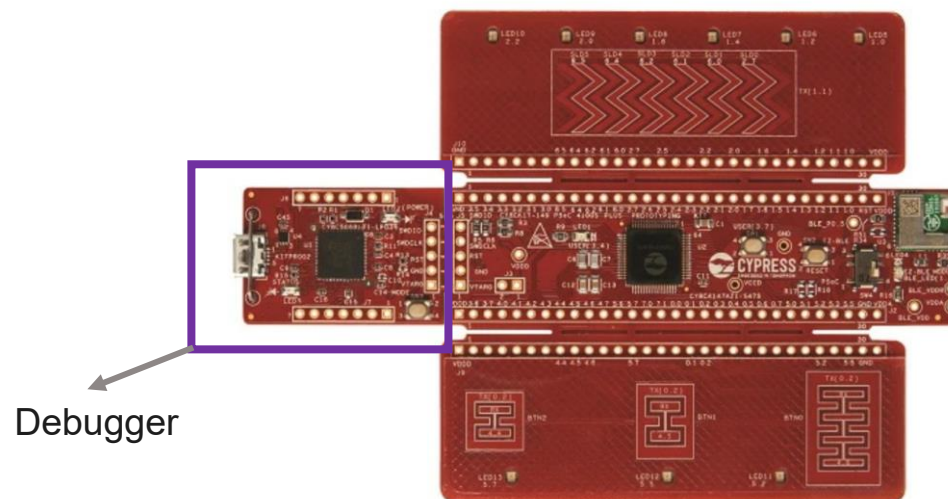
Host PC with cross-compiler for target CPU (e.g. ARM Cortex-M0)
Compilation, Linking and ELF generation

This debugger box helps in programming ELF sections of your application to Non-Volatile memories of the Microcontroller

When you power-up the CPU

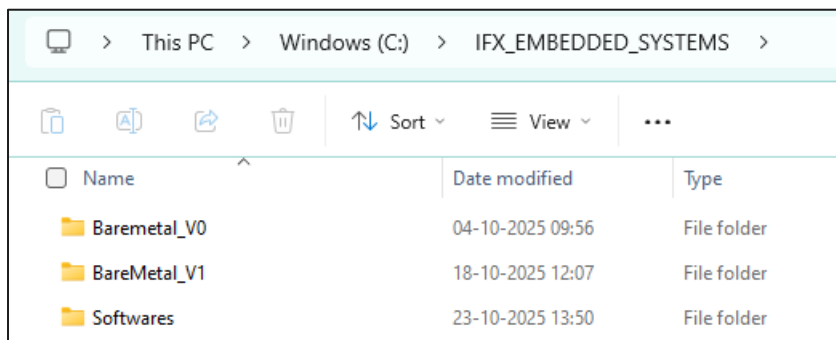
The startup software

1. Startup code
2. Goto main()



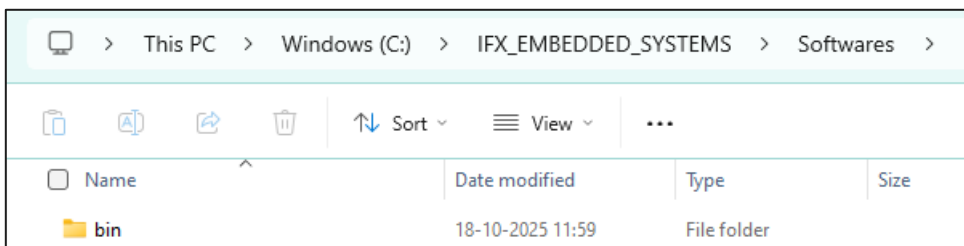
INSTALLATIONS 1/5

- 1 Verify that you had created a folder in **C:** with name **“IFX_EMBEDDED_SYSTEMS”**
And this folder should have following sub-folders:



Name	Date modified	Type
Baremetal_V0	04-10-2025 09:56	File folder
BareMetal_V1	18-10-2025 12:07	File folder
Softwares	23-10-2025 13:50	File folder

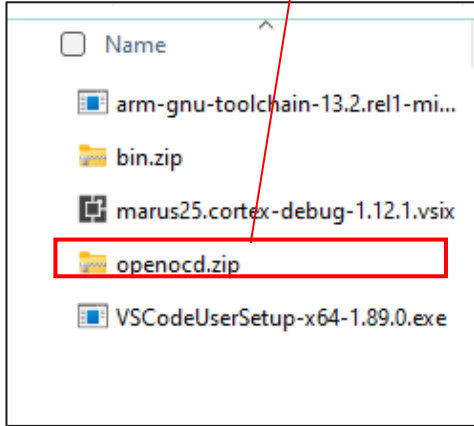
- 2 Verify the contents of Softwares folder in the path C:\IFX_EMBEDDED_SYSTEMS\Softwares



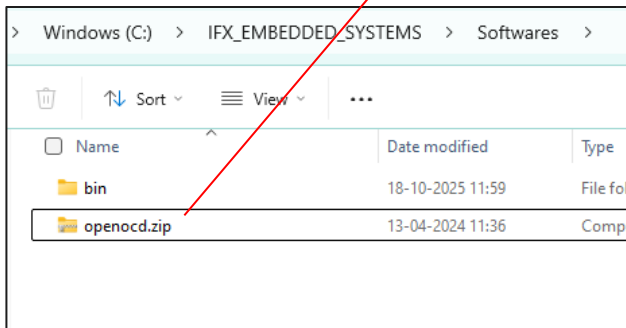
Name	Date modified	Type	Size
bin	18-10-2025 11:59	File folder	

INSTALLATIONS 2/5

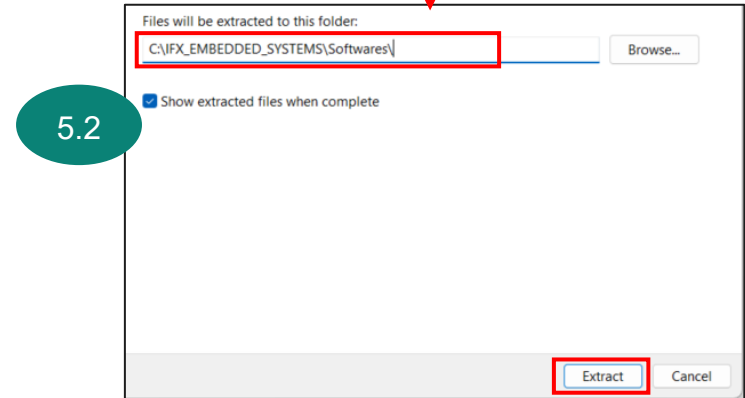
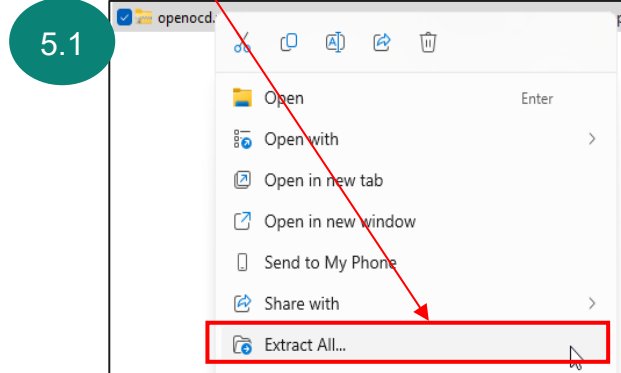
- 3 Copy the “openocd.zip” file from softwares.zip given by IFX in previous class



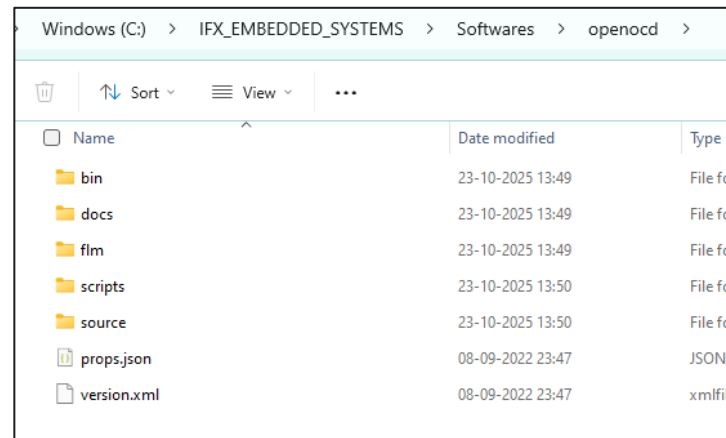
- 4 Paste the “openocd.zip” into path “C:\IFX_EMBEDDED_SYSTEMS\Softwares”



- 5 Unzip the “openocd.zip” at the same location (remember to remove “openocd” from extract path)

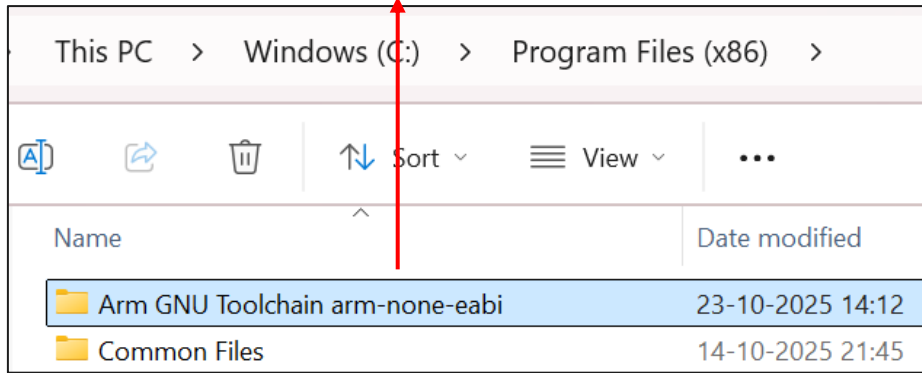


- 6 Verify the openocd folder in below path “C:\IFX_EMBEDDED_SYSTEMS\Softwares\openocd”

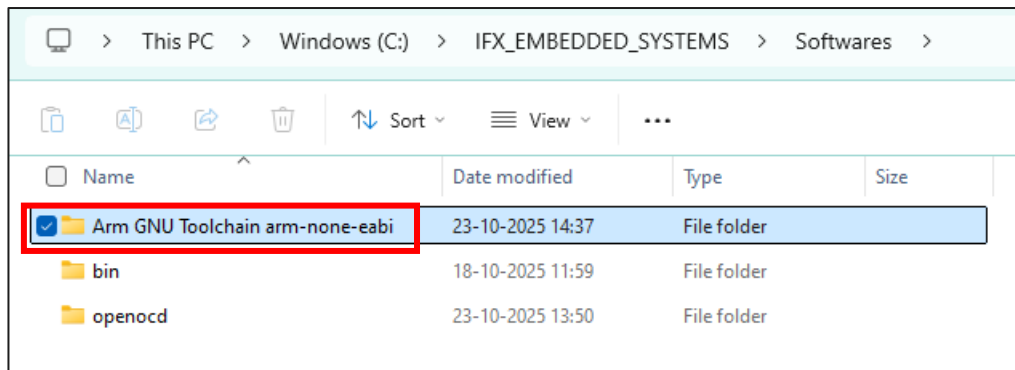


INSTALLATIONS 3/5

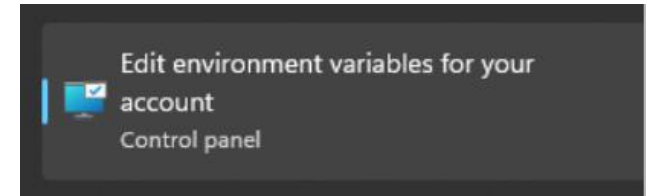
- 6 Go to your already installed arm-none-eabi tool location
(default location -> C:\Program Files (x86))
And **copy** the **Arm GNU Toolchain arm arm-none-eabi**



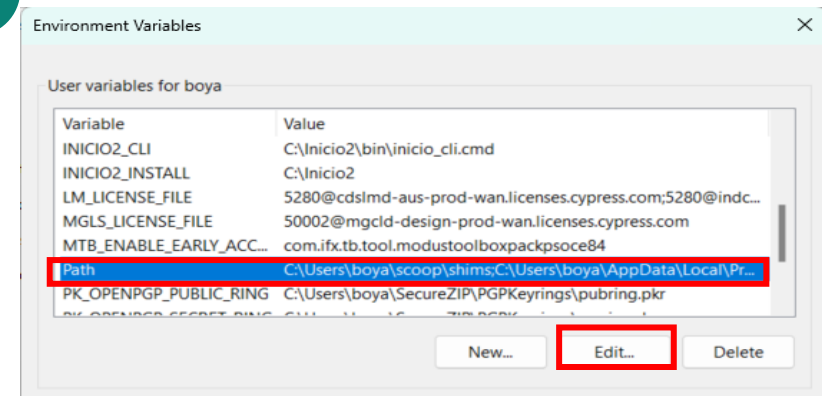
- 7 Paste the **Arm GNU Toolchain arm** folder into
C:\IFX_EMBEDDED_SYSTEMS\Softwares



- 8 Open Environmental variables for your
account from start menu



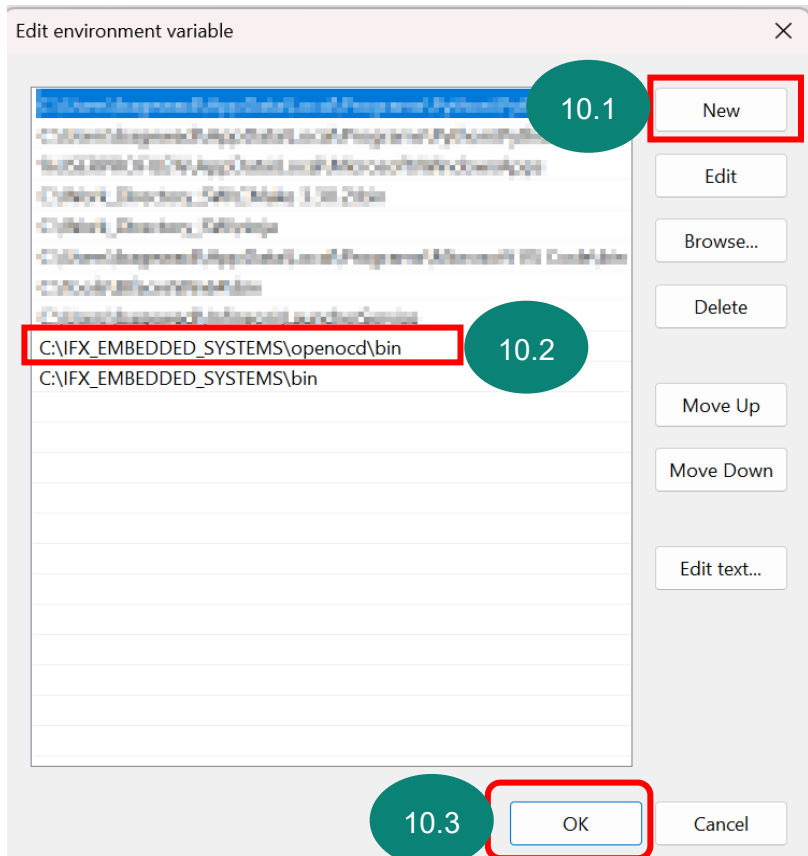
- 9 Click on Path and then Edit



INSTALLATIONS 4/5

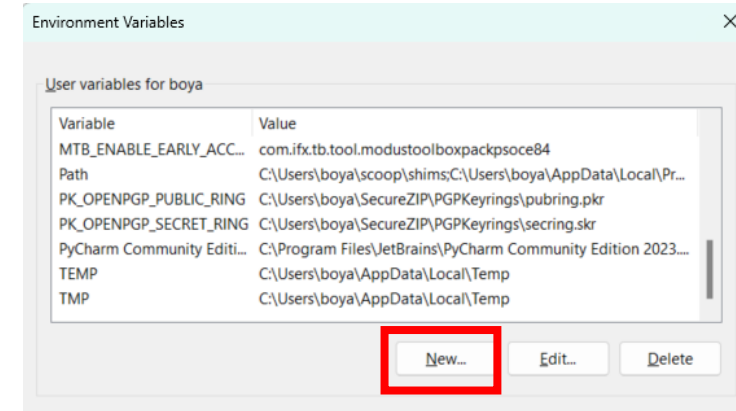
- 10 Add Openocd path using NEW button
(remember to add till openocd\bin)

C:\IFX_EMBEDDED_SYSTEMS\Softwares\openocd\bin

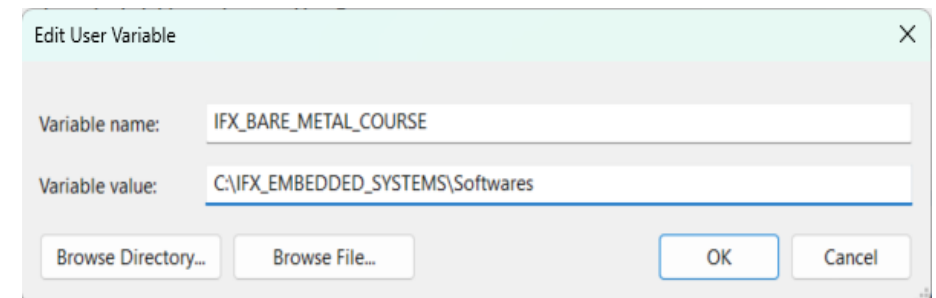


Click OK

- 11 Now again in Environment variables window, click on NEW button



- 12 Create new Variable as below:
Name: **IFX_BARE_METAL_COURSE**
Path: **C:\IFX_EMBEDDED_SYSTEMS\Softwares**



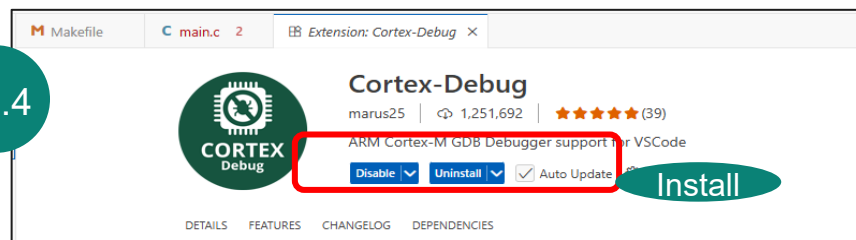
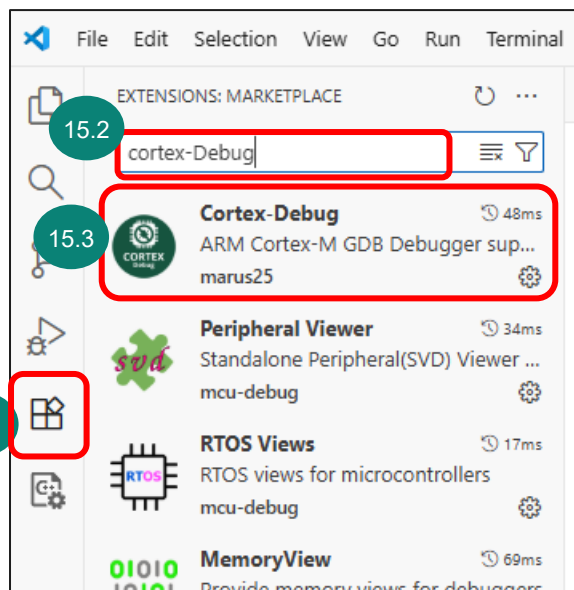
Steps for Debugging with VSCODE

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13. Open VSCode (preferably version 1.89.0 and above)

14. Go to FILE -> Open Folder -> select the Baremetal_V1 project folder path

15. Install “marus25.cortex-debug” extension from extensions window (CTRL + SHIFT + X).



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Go to RUN AND DEBUG -> select the launch config -> RUN

